

AZIM PREMJI UNIVERSITY, BHOPAL

First Semester 2026-2027

Tutorial Sheet-3

Course No. MTH-113

Course title: Introduction to Quantitative Thinking

- (1) What will be the unit digit of the squares of the following numbers?

(i) 81 (ii) 272 (iii) 799 (iv) 3853 (v) 1234 (vi) 26387

- (2) The squares of which of the following would be odd numbers?

(i) 431 (ii) 2826 (iii) 7779 (iv) 82004

- (3) Observe the following pattern and find the missing digits.

$$\begin{array}{rcl} 11^2 & = & 121 \\ 101^2 & = & 10201 \\ 1001^2 & = & 1002001 \\ 100001^2 & = & 1\ldots\ldots2\ldots\ldots1 \\ 10000001^2 & = & \ldots\ldots\ldots \end{array}$$

- (4) Using the given pattern, find the missing numbers.

$$\begin{array}{l} 1^2 + 2^2 + 2^2 = 3^2 \\ 3^2 + 4^2 + 12^2 = 13^2 \\ 4^2 + 5^2 + _ = 21^2 \\ 5^2 + _ + 30^2 = 31^2 \\ 6^2 + 7^2 + _ = _ \end{array}$$

- (5) How many numbers lie between squares of the following numbers?

(i) 12 and 13 (ii) 25 and 26 (iii) 99 and 100

- (6) Verify that the following equality holds for any integer $m \geq 0$:

$$(2m)^2 + (m^2 - 1)^2 = (m^2 + 1)^2$$

In this case, the triplet $(2m, m^2 - 1, m^2 + 1)$ is known as *Pythagorean triplet*.

- (7) Write a Pythagorean triplet whose smallest member is 8.

- (8) Write a Pythagorean triplet whose one member is:

(i) 6 (ii) 14 (iii) 16 (iv) 18

- (9) Find the smallest square number which is divisible by each of the numbers 6, 9 and 15.

- (10) What could be the possible 'one's' digits of the square root of each of the following numbers?

(i) 9801 (ii) 99856 (iii) 998001

- (11) Without doing any calculation, find the numbers which are surely not perfect squares.

(i) 153 (ii) 257 (iii) 408 (iv) 441

- (12) Find the square roots of the following numbers by the Prime Factorisation Method.

(i) 153 (ii) 1764 (iii) 4096 (iv) 7744

- (13) For each of the following numbers, find the smallest whole number by which it should be multiplied so as to get a perfect square number. Also find the square root of the square number so obtained.

(i) 252 (ii) 1008 (iii) 2028 (iv) 1458

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- (14) Find the square root of each of the following numbers by Division method.
 (i) 2304 (ii) 4489 (iii) 3481 (iv) 5776
- (15) Find the least number which must be subtracted from each of the following numbers so as to get a perfect square. Also find the square root of the perfect square so obtained.
 (i) 402 (ii) 1989 (iii) 3250 (iv) 825
- (16) Find the least number which must be added to each of the following numbers so as to get a perfect square. Also find the square root of the perfect square so obtained.
 (i) 525 (ii) 1750 (iii) 252 (iv) 1825
- (17) Find the smallest number by which each of the following numbers must be multiplied to obtain a perfect cube.
 (i) 243 (ii) 256 (iii) 72 (iv) 675
- (18) You are told that 1,331 is a perfect cube. Can you guess without factorisation what is its cube root? Similarly, guess the cube roots of 4913, 12167, 32768.
- (19) Find the value of.
 (i) $(3^0 + 4^{-1}) \times 2^2$ (ii) $(2^{-1} \times 4^{-1}) \div 2^{-2}$ (iii) $(\frac{1}{2})^{-2} + (\frac{1}{3})^{-2} + (\frac{1}{4})^{-2}$ (iv) $(3^{-1} + 4^{-1} + 5^{-1})^0$
 (v) $\frac{8^{-1} + 5^3}{2^{-4}}$ (vi) $\left\{ (\frac{1}{3})^{-1} - (\frac{1}{4})^{-1} \right\}^{-1}$
- (20) Express the following numbers in usual form.
 (i) 3.02×10^{-6} (ii) 4.5×10^4 (iii) 3×10^{-8} (iv) 3.61492×10^6
- (21) Express the number appearing in the following statements in standard form.
 (a) 1 micron is equal to $\frac{1}{1000000}$ m
 (b) Charge of an electron is 0.000,000,000,000,000,16 coulomb.
 (c) Size of a bacteria is 0.0000005 m
 (d) Size of a plant cell is 0.00001275 m
 (e) Thickness of a thick paper is 0.07 mm
